Statistical Analysis of Experimental Data

Homework 03

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https://github.com/igordt/SAED_homeworks/tree/main

Introduction

This homework addresses the evaluation of branching ratios: *consider an experiment measuring decays of the* $\omega(782)$ *meson, with three dominant decay channels considered:*

1. $\omega \to \pi^+\pi^-\pi^0$
2. $\omega \to \pi^0\gamma$
3. $\omega \to \pi^+\pi^-$

From $N = 4117$ *decays measured,* $N_1 = 3714$ *were classified as channel 1,* $N_2 = 338$ *as channel 2 and* $N_3 = 65$ *as channel 3. Calculate the corresponding branching ratios and their uncertainties.*

Below, I present the statistical model describing the measurement, the estimate of the branching ratios with their uncertainties, and a comparison between the correct and a commonly used wrong way of computing these uncertainties.

Statistical model

Each decay is a single trial whose outcome is one of the three channels, with probabilities p_1, p_2, p_3: these are the branching ratios we want to measure. Since we assume that every decay ends up in one of these three channels, the probabilities must satisfy the normalization condition

$$p_1 + p_2 + p_3 = 1. \tag{1}$$

Because each trial has more than two possible outcomes, the joint probability to observe N_1, N_2, N_3 decays in the three channels out of N independent decays is given by the multinomial distribution, the generalization of the binomial distribution:

$$P(N_1, N_2, N_3) = \frac{N!}{N_1!\, N_2!\, N_3!}\, p_1^{N_1}\, p_2^{N_2}\, p_3^{N_3}, \qquad N_1 + N_2 + N_3 = N. \tag{2}$$

As described in the PDG review, we can regard channel i as "success" and all the other channels as "failure". Then each N_i individually follows a binomial distribution with N tries, success probability p_i and failure probability $q_i = 1 - p_i$:

$$P(N_i) = \binom{N}{N_i} p_i^{N_i}\, q_i^{N-N_i}, \tag{3}$$

so that the expected value and the variance of each N_i are those of the binomial distribution:

$$\mathbb{E}(N_i) = p_i N, \qquad \mathbb{V}(N_i) = p_i(1 - p_i)N. \tag{4}$$

Note, however, that the N_i are not independent: since N is fixed, if one channel has more decays the others must have fewer. This will be relevant for the uncertainty estimates conducted below.

Branching ratios and their uncertainties

The branching ratio of channel i is estimated as the fraction of decays observed in that channel:

$$\hat{p}_i = \frac{N_i}{N}. \tag{5}$$

Using the binomial variance of Eq. (4), for fixed N, the variance of $\hat{p}_i$ is[1]

$$\sigma^2_{p_i} = \mathbb{V}\left(\frac{N_i}{N}\right) = \frac{\mathbb{V}(N_i)}{N^2} = \frac{p_i(1-p_i)\,N}{N^2} = \frac{\hat{p}_i\,(1-\hat{p}_i)}{N}, \tag{6}$$

where the unknown true p_i is replaced by its estimate $\hat{p}_i$. The resulting branching ratios are

$$\hat{p}_1 = 0.9021 \pm 0.0046, \qquad \hat{p}_2 = 0.0821 \pm 0.0043, \qquad \hat{p}_3 = 0.0158 \pm 0.0019, \tag{7}$$

and their sum is $\sum_i \hat{p}_i = 1$, as required by the normalization condition.

Poisson vs binomial approach

As discussed in the lecture, a common mistake is to compute the uncertainty of $\hat{p}_i = N_i/N$ by treating N_i and N as two independent variables, each with a Poisson uncertainty:

$$\sigma^2_{N_i} = N_i, \qquad \sigma^2_N = N. \tag{8}$$

Propagating the uncertainties for independent variables, $\sigma^2_{p_i} = \sum_k \left(\frac{\partial \hat{p}_i}{\partial x_k}\right)^2 \sigma^2_{x_k}$, with $x_k = N_i, N$ and

$$\frac{\partial \hat{p}_i}{\partial N_i} = \frac{1}{N}, \qquad \frac{\partial \hat{p}_i}{\partial N} = -\frac{N_i}{N^2}, \tag{9}$$

one obtains

$$\sigma^2_{p_i} = \left(\frac{1}{N}\right)^2 N_i + \left(-\frac{N_i}{N^2}\right)^2 N = \frac{N_i}{N^2} + \frac{N_i^2}{N^3} = \frac{\hat{p}_i\,(1+\hat{p}_i)}{N} \qquad \textbf{WRONG} \tag{10}$$

This result is wrong because N_i and N are not independent.

The correct approach is to write the branching ratio as

$$\hat{p}_i = \frac{N_i}{N_i + M_i}, \qquad M_i = N - N_i, \tag{11}$$

where M_i is the number of decays in all the other channels. Now N_i and M_i are two independent variables with Poisson uncertainty, $\sigma^2_{N_i} = N_i$ and $\sigma^2_{M_i} = M_i$. With the derivatives

$$\frac{\partial \hat{p}_i}{\partial N_i} = \frac{M_i}{(N_i + M_i)^2}, \qquad \frac{\partial \hat{p}_i}{\partial M_i} = -\frac{N_i}{(N_i + M_i)^2}, \tag{12}$$

and using $N_i + M_i = N$, the propagation gives

$$\sigma^2_{p_i} = \left(\frac{M_i}{N^2}\right)^2 N_i + \left(-\frac{N_i}{N^2}\right)^2 M_i = \frac{N_i\,M_i\,(M_i+N_i)}{N^4} = \frac{1}{N}\,\frac{N_i}{N}\,\frac{M_i}{N} = \frac{\hat{p}_i\,(1-\hat{p}_i)}{N}, \tag{13}$$

which is the same result obtained from the binomial distribution, Eq. (6). The numerical values of the three approaches are compared in Table 1.

[1] For a random variable X and a constant a, $\mathbb{V}(aX) = a^2\,\mathbb{V}(X)$. With $X = N_i$ and $a = 1/N$ the factor $1/N$ comes out squared.

Channel	$\hat{p}_i$	σ_{p_i} binomial	σ_{p_i} Poisson	σ_{p_i} Poisson - WRONG
1	0.9021	0.0046	0.0046	0.0204
2	0.0821	0.0043	0.0043	0.0046
3	0.0158	0.0019	0.0019	0.0020

Table 1: Uncertainties of the branching ratios obtained from the binomial distribution, from the correct Poisson approach and from the wrong Poisson approach.

The binomial approach and the correct Poisson approach give exactly the same uncertainties. The wrong approach overestimates the uncertainty.

Result

The branching ratios of the three dominant decay channels of the $\omega(782)$ meson, with their uncertainties obtained from the binomial distribution, are reported in Table 2.

Channel	BR	σ_{BR}	relative uncertainty
$\omega \to \pi^+\pi^-\pi^0$	0.9021	0.0046	0.5 %
$\omega \to \pi^0\gamma$	0.0821	0.0043	5.2 %
$\omega \to \pi^+\pi^-$	0.0158	0.0019	12.3 %

Table 2: Branching ratios of the $\omega(782)$ meson and their uncertainties.

The relative uncertainty grows as the number of decays in the channel decreases, so the rare channel $\omega \to \pi^+\pi^-$, with only 65 events, is the least precisely measured.